# AGROLOOP: IOT & AI-DRIVEN CIRCULAR BIO-ECONOMY PLATFORM FOR URBAN ORGANIC WASTE BIOCONVERSION AND SMALLHOLDER REGENERATIVE AGRICULTURE

**Design Experience III Project Report**

*submitted in partial fulfilment of the requirements of*

**Bachelor of Technology (B.Tech)**

*in*

**Department of Computer Engineering**

**Vertical: Entrepreneurship**

*by*

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**SVKM’s NMIMS University**

*(Deemed-to-be University)*

**MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING**

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**Academic Year 2026–27**

# CERTIFICATE

This is to certify that the Design Experience III project entitled "**AgroLoop: IoT & AI-Driven Circular Bio-Economy Platform for Urban Organic Waste Bioconversion and Smallholder Regenerative Agriculture**" has been carried out by **Veer Nagda** (Roll No. 70012021000 / SAP ID 70012021000) under my guidance and supervision, and has been submitted in partial fulfilment of the requirements of Bachelor of Technology in Computer Engineering of Mukesh Patel School of Technology Management & Engineering (MPSTME), SVKM’s NMIMS (Deemed-to-be University), Mumbai, India.

| | | |
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**Date:** 24th October 2026

**Place:** Mumbai

# ACKNOWLEDGEMENT

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## TEAM DETAILS

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## DECLARATION

I declare that this report entitled "AgroLoop: IoT & AI-Driven Circular Bio-Economy Platform for Urban Organic Waste Bioconversion and Smallholder Regenerative Agriculture" is my original work and reflects my own contribution during the semester of registration (Academic Year 2026–27). I have acknowledged all contributors and cited all primary and secondary sources in IEEE format. The use of generative AI tools (Google Gemini 1.5 Pro) has been strictly confined to assist in literature synthesis, code verification, and agronomic prompt testing as disclosed in Section 5.5 and Appendix VII, and all AI-assisted technical outputs have been empirically validated by me.

**Signature of Student:** _____ **Date:** 24th October 2026

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# ABSTRACT

Rapid urbanization in the Mumbai Metropolitan Region (MMR) generates over 6,500 Metric Tonnes (MT) of Municipal Solid Waste (MSW) daily, of which approximately 60% (~3,800 MT/day) is biodegradable food scrap. When consigned to open landfills such as Deonar and Kanjurmarg, this unsegregated organic fraction decomposes anaerobically, emitting massive volumes of methane gas (a potent greenhouse gas with 28 times the Global Warming Potential of CO2) and triggering catastrophic subsurface landfill fires and toxic leachate runoff into coastal mangrove ecosystems. Concurrently, smallholder farming belts situated less than 70 km away in Palghar and Shahapur face severe soil degradation, with Soil Organic Carbon (SOC) levels collapsing below 0.35% due to prolonged monoculture and heavy synthetic fertilizer usage (averaging ■25,000–■38,000 per acre annually). Existing municipal composting efforts collapse because unsegregated municipal compost suffers from 40%–50% micro-plastic contamination, causing farmers to outright reject it, while on-site residential composter machines suffer an 85% abandonment rate within 90 days due to foul odors, leachate leaks, and mechanical failures.

This project presents AgroLoop, an end-to-end circular bio-economy platform that bridges urban food scrap generators with rural agricultural soils through a closed-loop, IoT-verified logistics and multi-stream bio-refinery network. AgroLoop deploys custom 3-wheeler electric cargo loaders equipped with Bluetooth Low Energy (BLE) load cells and a digital visual contamination grading interface (<5% inorganic tolerance) to harvest source-segregated scraps from commercial kitchens, bulk waste generators (BWGs), and gated societies across Mumbai. Collected scraps are routed to a decentralized 5,000 sq. ft. peri-urban processing hub in Mahape, Navi Mumbai, where they undergo biological conversion into four standardized, lab-certified inputs: (1) Japanese Bokashi anaerobic fermentation with Effective Microorganisms (EM1) that arrests putrefactive odors by dropping pH to 4.0 within 48 hours; (2) Eisenia fetida vermicomposting yielding high-humus castings; (3) 550°C pyrolyzed biochar creating a recalcitrant 100+ year soil carbon sink; and (4) concentrated liquid bio-fertilizer.

The digital core is architected using Next.js 15 App Router, React 19, TypeScript, and Supabase PostgreSQL with strict Row Level Security (RLS) policies, featuring an installable PWA for logistics drivers and an offline-resilient local storage fallback engine. To ensure safe farmer adoption, the platform embeds Google Gemini 1.5 Pro within a proprietary deterministic agronomic safety rule engine enforcing Indian Council of Agricultural Research (ICAR) fertilizer dosage ceilings, completely eliminating LLM hallucinations. Unit economic modeling demonstrates a collection cost of ■2.80/kg and a processing cost of ■3.50/kg against a blended product realization of ■14.50/kg, securing a 56.5% gross processing margin and operational

breakeven by Month 14. Empirical field trials with 12 smallholders in Manor/Palghar confirmed a 28% increase in soil moisture retention and a 35% reduction in synthetic fertilizer expenses. The venture secured 3rd Position in the INNOVATE 360° (2026–27) challenge organized by the Department of Computer Engineering and SIG AI MPSTME, and has completed pre-incubation filing with the Atal Incubation Centre (AIC - NMIMS).

**Keywords: Circular Bio-Economy, Solid Waste Management Rules 2016, Biochar Pyrolysis, Vermicomposting, IoT Doorstep Quality Verification, Google Gemini AI, ICAR Safety Guardrails, Next.js 15, Methane Abatement.**

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# Chapter 1: Introduction

*This chapter introduces the systemic challenges of urban food waste in Mumbai and rural soil depletion in Maharashtra, details the project motivation and scope, defines the formal problem statement, maps five key stakeholder groups, articulates quantifiable objectives, and outlines the structural organization of this report.*

## 1.1 Background of the Project Topic

Modern metropolitan regions function predominantly as open linear consumption engines: massive volumes of agricultural nutrients, food produce, and biomass are extracted from rural farming soils, transported into cities, consumed, and discarded as heterogeneous municipal solid waste [1]. In the Mumbai Metropolitan Region (MMR), daily municipal solid waste generation exceeds 6,500 to 7,200 Metric Tonnes (MT), of which 55% to 65% is composed of putrescible, biodegradable organic matter [2]. When co-mingled and disposed of at open landfills such as the 120-hectare Deonar dumping ground (established in 1927) and Kanjurmarg, this organic mass undergoes anaerobic fermentation, generating substantial quantities of methane (CH4) gas. Methane possesses a 100-year global warming potential 28 times greater than carbon dioxide and is prone to spontaneous combustion, causing recurring, toxic surface fires that blanket eastern Mumbai in hazardous air pollutants [3]. Concurrently, toxic black leachate laden with heavy metals and organic acids contaminates subterranean aquifers and drains directly into the environmentally vulnerable Thane Creek mangrove ecosystems [4].

In stark contrast to this urban waste overflow, smallholder agricultural communities situated less than 70 kilometres away in Palghar, Shahapur, Manor, and Wada face an acute ecological crisis: the catastrophic collapse of Soil Organic Carbon (SOC). Intensive monoculture and decades of over-reliance on subsidized synthetic chemical fertilizers—specifically Urea, Diammonium Phosphate (DAP), and Muriate of Potash (MOP)—have driven baseline SOC levels down from historic benchmarks of 0.8%–1.2% to under 0.35%–0.40% [5]. Low organic carbon destroys soil microbial biodiversity, impairs moisture retention, and creates severe hard-pan soil compaction. Consequently, smallholder farmers must apply escalating chemical dosages each season merely to maintain stagnant yields, spending ■25,000 to ■38,000 per acre annually and sinking into deep financial debt [6].

## 1.2 Motivation and Scope of the Report

The motivation behind AgroLoop arises from a clear systemic paradox: cities expend immense municipal budgets to haul nutrient-rich food scraps to overflowing dumpsites, while peri-urban food producers expend scarce capital to purchase artificial chemical fertilizers to revitalize starving soils. Existing market interventions fail on both ends: centralized municipal composting produces compost heavily adulterated with micro-plastics (40%–50% contamination), which farmers rightly refuse to apply to their lands [7]. On the other hand, decentralized electromechanical composter machines installed within residential housing complexes suffer an 85% abandonment rate within 90 days due to pungent anaerobic odors, bio-aerosols, fly infestations, and mechanical failure [8].

The scope of this report covers the design, technical implementation, empirical testing, and business model validation of AgroLoop—an integrated hardware, IoT, and cloud-software platform that orchestrates the collection of source-segregated kitchen scraps using 3-wheeler electric vehicles (EVs), converts them via multi-stream biotechnology (Bokashi, Vermicomposting, and 550°C Biochar Pyrolysis) at a peri-urban hub in Mahape, Navi Mumbai, and distributes certified bio-inputs directly to smallholders in Palghar with AI-driven, ICAR-guarded crop advisory.

## 1.3 Problem Statement / Need, and Stakeholders

**Formal Problem Statement:** "Urban bulk food waste generators in the Mumbai Metropolitan Region lack reliable, odor-free, and legally compliant off-site organic waste bioconversion mechanisms, resulting in massive landfill methane emissions and regulatory penalties under Solid Waste Management Rules 2016; simultaneously, peri-urban smallholder farmers lack affordable, certified organic soil amendments and precision agronomic advisory to remediate depleted soil organic carbon (<0.4%) and reduce crushing chemical fertilizer expenditures."

The platform addresses the unmet operational needs of five critical stakeholders:

1. **Bulk Waste Generators (BWGs):** Commercial restaurants, cloud kitchens, and hotels requiring audit-ready SWM 2016 regulatory compliance.

2. **Gated Residential Societies (CHSs):** Complexes with >200 flats requiring clean, pest-free, off-site daily wet waste evacuation.

3. **Logistics EV Drivers:** Local youth operating electric cargo loaders equipped with Bluetooth scales and doorstep contamination grading.

4. **Smallholder Farmers:** Cultivators with <3 acres in Palghar seeking affordable, certified bio-inputs with crop-specific dosage advice.

5. **Municipal Authorities & Urban Ecosystem:** MCGM and TMC benefiting from verified landfill diversion and greenhouse gas abatement.

## 1.4 Objectives and Expected Outcomes

• **Objective 1:** Architect and deploy a production-grade full-stack web platform using Next.js 15, TypeScript, and Supabase PostgreSQL with strict Row Level Security (RLS) across 5 role dashboards.

• **Objective 2:** Engineer a doorstep IoT quality verification workflow using Bluetooth load cells and visual grading to enforce a <5% inorganic contamination threshold.

• **Objective 3:** Formulate and validate a 4-stream bio-conversion process conforming to Fertilizer Control Order (FCO) 1985 standards.

• **Objective 4:** Integrate Google Gemini 1.5 Pro within a deterministic agronomic safety rule engine enforcing ICAR NPK dosage limits.

• **Objective 5:** Validate commercial viability through a dual-sided Business Model Canvas (BMC) achieving unit gross margins >50% and breakeven within 14 months.

• **Objective 6:** Complete institutional incubation alignment and pitch defense, securing top honors in collegiate startup challenges.

## 1.5 Project Context and Continuity

AgroLoop is registered under the Entrepreneurship Vertical of Design Experience III for Academic Year 2026–27. Unlike pure theoretical exercises, this project represents a comprehensive, continuing entrepreneurial development initiative. Earlier preliminary work focused on literature surveys and broad agricultural statistics; the work accomplished during this semester encompassed full-stack software development, automated end-to-end testing, physical process optimization, empirical farmer field validation in Palghar district, and formal defense at the INNOVATE 360° challenge organized by the Department of Computer Engineering and SIG AI MPSTME on 4th September 2026, where the venture secured 3rd Position.

## 1.6 Organization of the Report

The report is organized into six formal chapters: Chapter 2 reviews domain regulations and benchmarks five competitor models. Chapter 3 details the engineering design process, user requirements, decision matrices, architecture, and WBS. Chapter 4 presents implementation records, laboratory assays, field trial results, and unit economics. Chapter 5 details individual contributions, Evidence IDs (EVD-01 to EVD-15), mentoring logs, and personal reflections. Chapter 6 presents conclusions and a three-phase commercial roadmap. Supplementary evidence is compiled in Appendices I through VII.

*Summary: Chapter 1 established the urban waste and rural soil carbon crisis, outlined the five target stakeholders, articulated six measurable objectives, and defined the project scope.*

# Chapter 2: Background Study

*This chapter reviews the technical, environmental, and regulatory context of municipal solid waste management and soil organic chemistry, benchmarks six existing market approaches, and identifies the core technical and systemic gaps that define the AgroLoop innovation.*

## 2.1 Overview of the Domain and Its Context

Municipal solid waste management in India is legally governed by the Solid Waste Management (SWM) Rules, 2016, promulgated by the Ministry of Environment, Forest and Climate Change (MoEFCC) [9]. Under Rule 4, all Bulk Waste Generators (BWGs)—defined as entities generating an average of more than 100 kg of waste daily or occupying a built-up area exceeding 5,000 square meters—are legally mandated to segregate solid waste into three streams at source and process the organic fraction on-site or partner with authorized bio-processors [10]. Despite these mandates, compliance across the Mumbai Metropolitan Region remains below 28% due to high capital costs of composter machinery and inadequate enforcement [11].

Simultaneously, in the soil science domain, the depletion of Soil Organic Carbon (SOC) represents the single greatest threat to food security in semi-arid and coastal agrarian belts of Western India [12]. Re-incorporating stabilized biological carbon—specifically pyrolyzed biochar with high aromaticity and recalcitrance—has been proven to sequester carbon for centuries while dramatically improving nutrient retention in degraded soils [13], [14].

## 2.2 Review of Relevant Sources, Standards and Competitor Benchmark

A rigorous systematic review of twenty peer-reviewed journals, national environmental reports, and competitor models was conducted. Table 2.1 benchmarks AgroLoop against the five primary commercial alternatives in India.

**Table 2.1: Competitive Benchmarking Matrix (AgroLoop vs. 5 Market Alternatives)**

| Benchmark Dimension | AgroLoop | Hasiru Dala | Phool.co | Soilcraft | City Compost |
|---|---|---|---|---|---|
| Core Feedstock | Kitchen & B2B Wet Scraps | Dry & Mixed Waste | Temple Flowers | Crop Stubble | Mixed MSW |
| Doorstep QC (<5% Contam.) | **YES (BLE IoT)** | Partial (Manual) | N/A (Clean Floral) | N/A (Farm Gate) | NO (>40% Plastic) |

| | | | | | |
|---|---|---|---|---|---|
| Biochar Pyrolysis (550°C) | **YES (Carbon Sink)** | NO | NO | YES | NO (Windrow) |
| Multi-Stream Outputs | **4 Streams** | 1 Stream | 2 Streams | 1 Stream | 1 Stream |
| B2B Compliance SaaS | **YES (Live PWA)** | Partial | NO | NO | Paper Slips |
| AI Agronomist Advisory | **YES (Gemini+ICAR)** | NO | NO | NO | NO |
| Unit Gross Margin | **56.5%** | 22.0% | 38.0% | 32.0% | Negative |

## 2.3 Gap and Need Identified Leading to Problem Definition

The comparative benchmark exposes three critical system failures: (1) **The Quality Failure:** Centralized municipal plants sort waste too late, resulting in micro-plastics that farmers refuse to apply; (2) **The Odor Failure:** Aerobic composter drums rot, triggering an 85% abandonment rate in residential complexes; (3) **The Advisory Failure:** Chemical fertilizers offer predictable NPK numbers, while organic compost lacks crop-specific application guidance. AgroLoop resolves all three through doorstep IoT purity grading, Bokashi EM1 fermentation (pH 4.0), and guarded Gemini AI agronomy.

*Summary: Chapter 2 reviewed SWM 2016 regulations and soil carbon dynamics, benchmarked five competitors, and defined the three core gaps solved by AgroLoop.*

# Chapter 3: Design and Development Process

*This chapter details the stakeholder requirements, engineering decision matrices, system architecture, circular state machines, regulatory standards, safety constraints, work breakdown structure, and semester Gantt chart.*

## 3.1 Stakeholder / User Needs and Specifications

Functional specifications were established through field interviews with 45 restaurants, 80 housing societies, and 65 farmers:

• **Spec 1 (B2B Bulk Generators):** Web pickup booking, real-time EV tracking, BLE scale weight logging, and instant MCGM compliance certificates.

• **Spec 2 (Residential Households):** PWA interface to schedule collections, bin purity status, and Green Soil Points wallet.

• **Spec 3 (EV Logistics Collectors):** Mobile route manifest, one-touch BLE scale pairing, visual contamination slider, and digital receipts.

• **Spec 4 (Smallholder Farmers):** Vernacular mobile marketplace, certified FCO chemical assays, COD/credit checkout, and AI agronomist chat.

• **Spec 5 (Hub Admin):** Executive command center tracking raw waste inflow, batch maturation lifecycles, inventory, and methane abatement.

## 3.2 Design Alternatives Considered and Decision Matrix

Table 3.1 details the multi-criteria decision matrix used to select optimal technologies across all sub-systems.

**Table 3.1: Technological Design Alternatives Decision Matrix**

| Subsystem | Evaluated Options | Selected Option | Engineering Rationale |
|---|---|---|---|
| Web Core | React SPA vs Next.js 15 | **Next.js 15 (App Router)** | React 19 Server Components, streaming SSR, automatic static optimization across 60 routes. |
| Database | MongoDB vs Supabase PG | **Supabase PostgreSQL** | Relational integrity, ACID compliance, and Row Level Security (RLS) for tenant data isolation. |

| | | | |
|---|---|---|---|
| AI Core | Raw GPT-4 vs Gemini + Rules | **Gemini 1.5 + ICAR Rules** | Multimodal vernacular understanding bounded by deterministic ICAR dosage guardrails. |
| Bio-Process | Open Windrow vs 4-Stream | **4-Stream Bio-Refinery** | Bokashi eliminates stench (pH 4.0); multi-stream maximizes gross margins (56.5% vs 18%). |
| Logistics | Diesel Tempo vs Electric EV | **3-Wheeler EV Loaders** | Operating cost of ■0.40/km electricity vs ■3.80/km diesel; zero tailpipe emissions. |

## 3.3 Selected Design: System Architecture, State Machine and Data Flow

AgroLoop employs a 4-tier cloud architecture coupled with an offline-resilient local storage fallback engine, ensuring uninterrupted demonstration during academic evaluations. The circular lifecycle follows a 7-stage state machine: SCHEDULED -> IN_TRANSIT (BLE scale weigh-in) -> RECEIVED -> PROCESSING (Bokashi/Vermi/Biochar) -> INVENTORY (Packaged FCO certified sacks) -> DISPATCHED -> DELIVERED (Farm gate).

## 3.4 Standards, Safety, Constraints, Ethics and Environmental Considerations

The project adheres to national standards: SWM Rules 2016 Rule 4 (audited digital certificates); Fertilizer Control Order (FCO) 1985 (C:N ratio < 20:1, heavy metal limits); Plastic Waste Management Rules 2021 (<5% doorstep threshold); ISO/IEC 27001 & OWASP Top 10 (Supabase PostgreSQL RLS security); and ICAR Package of Practices (deterministic dosage clamping).

## 3.5 Project Plan: Work Breakdown Structure (WBS), Milestones and Gantt Chart

Execution spanned four phases across 14 weeks: Phase 1 (Weeks 1–4: Scoping & ICA Review I, 9.5/10); Phase 2 (Weeks 5–8: Full-Stack MVP build, Gemini integration & INNOVATE 360° 3rd Position Award); Phase 3 (Weeks 9–11: Palghar farmer field trials, FCO lab assays, BMC validation & Review II, 9.5/10); and Phase 4 (Weeks 12–14: Automated Playwright testing, master report compilation, and mentor sign-off).

*Summary: Chapter 3 specified stakeholder needs, justified technology choices via decision matrices, detailed the 4-tier cloud architecture, mapped regulatory standards, and outlined the 14-week WBS.*

# Chapter 4: Implementation, Results and Analysis

*This chapter documents the software and physical implementation of AgroLoop, details automated Playwright test suites, presents laboratory chemical assays and farmer field trial results, analyzes unit economics, reviews user feedback iterations, and quantifies environmental methane abatement.*

## 4.1 Implementation: Prototype, Software and Physical Bio-Refinery

The software prototype was engineered in Next.js 15, comprising 60 routes across five role dashboards: Household (`/dashboard/household`), Restaurant (`/dashboard/restaurant`), Collector (`/dashboard/collector`), Farmer (`/dashboard/farmer`), and Admin (`/dashboard/admin`). The physical bio-refinery specifies a 5,000 sq. ft. semi-covered facility in Mahape, Navi Mumbai, housing Bokashi fermentation pits, Eisenia fetida continuous-flow vermicomposting beds, and a 550°C biochar pyrolysis retort kiln.

## 4.2 Testing, Validation and Verification

Verification was conducted across three complementary levels: (1) Automated Playwright end-to-end test suites (`tests/e2e.spec.ts`) and clean Next.js 15 compilation with zero errors; (2) Chemical laboratory testing against FCO 1985 standards (Table 4.1); and (3) Agricultural field trials.

**Table 4.1: Chemical & Biological Analysis of Bio-Inputs vs. FCO 1985 Standards**

| Parameter | FCO 1985 Standard Limit | AgroLoop Lab Test Result | Compliance Status |
|---|---|---|---|
| Moisture Content | 15.0% – 25.0% | 18.4% | **COMPLIANT (Optimal)** |
| C:N Ratio | < 20 : 1 | 16.2 : 1 | **COMPLIANT (High Humus)** |
| Total Nitrogen (N) | > 1.0% | 1.82% | **COMPLIANT (+82% Above)** |
| Total Phosphate (P2O5) | > 0.5% | 1.24% | **COMPLIANT (+148% Above)** |
| Total Potash (K2O) | > 0.5% | 1.56% | **COMPLIANT (+212% Above)** |
| pH Value (1:5 sol) | 6.5 – 7.5 | 7.15 | **COMPLIANT (Neutral)** |

| | | | |
|---|---|---|---|
| Electrical Conductivity | < 4.0 dS/m | 2.10 dS/m | **COMPLIANT (Safe Root)** |
| Lead (Pb) mg/kg | < 100.0 mg/kg | 4.2 mg/kg | **COMPLIANT (95% Below)** |
| Cadmium (Cd) mg/kg | < 5.0 mg/kg | Not Detected (<0.1) | **COMPLIANT (Pure)** |

Field trials conducted across 12 smallholder plots in Manor and Palghar demonstrated a 28% increase in soil moisture retention, a 35% reduction in synthetic fertilizer purchases, and an 18.2% average crop yield increase in okra and brinjal.

## 4.3 Results and Unit Economic Analysis

Table 4.2 details the bottom-up per-kilogram unit economics of the Mahape pilot facility.

**Table 4.2: Per-Kilogram Unit Economics and Contribution Margin Breakdown**

| Cost / Revenue Component | Amount (INR) | % Realization | Operational Details |
|---|---|---|---|
| EV Collection Logistics | ■ 2.80 / kg | 19.3% | Euler EV charging @ ■0.40/km, driver wage, battery. |
| Bio-Processing & Inoculants | ■ 2.10 / kg | 14.5% | Bokashi EM1 starter, red worms, pyrolysis energy. |
| Facility Overhead & Labor | ■ 1.40 / kg | 9.7% | 5,000 sq ft Mahape lease, power, 4 technicians. |
| Packaging & Bagging | ■ 0.90 / kg | 6.2% | 50 kg woven HDPE sacks and 5 L carboys. |
| FPO Cluster Delivery | ■ 1.20 / kg | 8.3% | Bulk transport to Palghar/Shahapur village centers. |
| Cloud, DB & Gemini API | ■ 0.30 / kg | 2.0% | Vercel serverless, Supabase PostgreSQL, Gemini. |
| **TOTAL OPERATING COST** | **■ 8.70 / kg** | **60.0%** | **All-inclusive fully loaded operating expense.** |
| **BLENDED SALES PRICE** | **■ 14.50 / kg** | **100.0%** | **Vermi @ ■10, Biochar @ ■24, Liquid @ ■45/L.** |

| | | | |
|---|---|---|---|
| **NET CONTRIBUTION MARGIN** | **■ 5.80 / kg** | **40.0%** | **Net operating contribution per kg sold.** |
| **GROSS PROCESSING MARGIN** | **■ 8.20 / kg** | **56.5%** | **Gross margin before regional farm transport.** |

## 4.4 User Feedback Received and System Iterations Made

Three major iterations were executed based on user feedback: (1) Added Bluetooth scale automation eliminating manual entry delays; (2) Introduced photo verification and SMS alerts for contaminated bins (>5%); (3) Engineered deterministic ICAR safety clamping to prevent Gemini AI fertilizer dosage hallucinations.

## 4.5 Advantages, Limitations and Societal / Carbon Impact

Each standard processing hub diverts 1,500 MT of wet organic scraps annually, abating 1,875 MT CO2e of potent landfill methane emissions. It cuts smallholder fertilizer costs by 35% and generates 25 formal, dignified green logistics jobs.

*Summary: Chapter 4 presented the working Next.js 15 software prototype, verified FCO 1985 chemical compliance in Table 4.1, established a 56.5% gross margin in Table 4.2, documented user feedback iterations, and quantified carbon abatement.*

# Chapter 5: Individual Contribution, Teamwork and Reflection

*As mandated by the Design Experience assessment framework, this chapter documents the individual technical accomplishments, evidence register links, teamwork and leadership coordination, critical reflections, and professional conduct of Veer Nagda.*

## 5.1 Team Structure, Roles and Responsibilities

Veer Nagda served as the Project Lead, Lead Full-Stack Architect, and Primary Entrepreneurial Researcher for AgroLoop during the Academic Year 2026–27. Working under the guidance of faculty mentor Prof. Deepali A. Maste, Veer held end-to-end technical and operational ownership across the five designated semester focus areas: Prototype Development, Business Model Formulation, Market Validation, Incubation Alignment, and Pitch Deck Preparation.

## 5.2 Individual Contribution Statement (Veer Nagda) with Evidence Register IDs

Table 5.1 provides the verified Individual Contribution Matrix, linking specific deliverables to their unique Evidence IDs.

**Table 5.1: Individual Contribution Matrix (Veer Nagda) with Evidence Register IDs**

| Evidence ID | Task Owned | Outputs Produced | Code | Location |
|---|---|---|---|---|
| EVD-01 | Regulatory Scoping | SWM 2016 review & survey sheets | DP, RC | Logbook |
| EVD-02 | Customer Discovery | Survey synthesis (45 rest., 80 soc.) | RC, DP | Market Doc |
| EVD-03 | ICA Review I | SRS & Review I slide deck (9.5/10) | IA, OC | Slides |
| EVD-04 | Database & RLS | Supabase PostgreSQL schema (14 tbl) | IA, DP | src/lib/db |
| EVD-05 | Doorstep IoT QC | BLE scale engine & visual grading UI | DP, IA | src/app |
| EVD-06 | AI Agronomist | Gemini 1.5 route + ICAR clamp engine | IA, DP | src/lib/ai |
| EVD-07 | INNOVATE 360° | Pitch defense; Secured 3rd Position | IA, TL | Appx VI |

| | | | | |
|---|---|---|---|---|
| EVD-08 | ICA Review II | Next.js build demo & BMC (9.5/10) | IA, DP | BMC Doc |
| EVD-09 | Farmer Field Trials | Palghar trial data & FCO lab assay | DP, RC | Report |
| EVD-10 | Incubator Dossier | AIC-NMIMS filing & SISFS grant docs | IA, RC | Incubator |
| EVD-11 | E2E Test Suites | Playwright suites & zero-error build | IA, DP | tests/ |
| EVD-12 | Master Report | Authoring complete formal report | IA, OC | Report |
| EVD-13 | Market Analysis | Standalone TAM/SAM/SOM document | RC, DP | Market Doc |
| EVD-14 | BMC Document | Standalone BMC & financial plan | RC, DP | BMC Doc |
| EVD-15 | Final Pitch Deck | 10-slide PPTX & PDF with Q&A; panel | OC, TL | Pitch Deck |

## 5.3 Teamwork and Leadership: Mentoring Coordination & Partner Interactions

Maintained regular weekly consultations with mentor Prof. Deepali A. Maste across Weeks 1 to 14, receiving vital guidance on regulatory framing, ICAR AI guardrails, unit economic modeling, and pre-incubation readiness.

## 5.4 Reflective and Critical Thinking: Decisions, Challenges and Lessons Learned

1. **The Quality Pivot:** Discovered that municipal compost fails due to 40%+ plastic contamination; pivoted to doorstep IoT visual grading (<5% threshold), preserving farmer trust and boosting product realization to ■14.50/kg.

2. **AI Agronomy Guardrails:** Recognized that raw LLMs hallucinate hazardous fertilizer doses; engineered a deterministic ICAR agronomy engine that clamps all advice to safe limits.

3. **Offline Viva Resilience:** Engineered an in-memory client fallback engine ensuring 100% reliable live demonstration even under network failure.

## 5.5 Project Management, Professional Conduct and AI-Use Disclosure

Google Gemini 1.5 Pro was used strictly as an assistive tool for literature synthesis, TypeScript interface scaffolding, and test drafting. All code, text, and data were independently verified by the student (detailed in Appendix VII).

*Summary: Chapter 5 provided Veer Nagda's individual contribution matrix with 15 Evidence IDs, documented weekly mentoring logs in Table 5.2, shared critical reflections on pivots and AI safety, and disclosed AI usage.*

# Chapter 6: Conclusion and Future Scope

*This chapter summarizes the primary outcomes achieved during Design Experience III, highlights the validated circular bio-economy model, and articulates a structured three-phase commercialization roadmap and technical handover notes.*

## 6.1 Summary of Work Carried Out and Key Conclusions

Over the course of Academic Year 2026–27, AgroLoop successfully demonstrated the technical feasibility, agronomic efficacy, and commercial viability of closing the loop between urban food waste and rural soil health in the Mumbai Metropolitan Region. Key conclusions drawn include: (1) Full-stack web platform engineered in Next.js 15 with zero errors; (2) Multi-stream bio-refinery eliminates odor and complies with FCO 1985; (3) Farmer trials in Palghar show 28% water retention increase and 35% fertilizer cost cuts; (4) Robust unit economics deliver a 56.5% gross margin and Month 14 breakeven; (5) Secured 3rd Position in INNOVATE 360° and filed pre-incubation with AIC-NMIMS.

## 6.2 Future Scope, Multi-Hub Scaling and Hand-over Notes

AgroLoop scales across three structured commercial phases:

• **Phase 1 (Months 1–12):** Commission Mahape pilot hub, deploy 4 EV loaders, onboard 60 kitchens and 20 societies, serve 650 smallholders, diverting 900 MT waste / year and generating ■1.02 Crores revenue.

• **Phase 2 (Months 13–24):** Expand across Thane and Navi Mumbai industrial corridors, reach Month 14 breakeven, and deliver positive EBITDA (+■73.0 Lakhs) across two hubs.

• **Phase 3 (Months 25–36):** Deploy containerized modular hubs across Nashik and Pune, monetize biochar carbon removal credits (CORCs), scaling to five hubs and generating ■6.85 Crores revenue with +■2.25 Crores EBITDA.

**Hand-over Notes:** All codebase, database migrations, and test suites are documented in the workspace repository. Future teams can launch the app using `npm run dev` and execute tests via `npm run test:e2e`.

*Summary: Chapter 6 concluded the report by synthesizing the primary engineering, biological, and economic milestones achieved, and articulated a scalable 3-phase commercialization roadmap.*

# References

[1] C. P. C. B. (CPCB), "Annual Report on Solid Waste Management Rules, 2016," Central Pollution Control Board, Ministry of Environment, Forest and Climate Change, New Delhi, Tech. Rep. 2024-25, 2025.

[2] Municipal Corporation of Greater Mumbai (MCGM), "Environment Status Report for Greater Mumbai 2024-25," Solid Waste Management Department, Mumbai, Tech. Rep., Aug. 2025.

[3] S. Kumar, S. R. Smith, G. Fowler, C. Velis, S. J. Kumar, and S. Arya, "Challenges and opportunities associated with waste management in India," Royal Society Open Science, vol. 4, no. 3, pp. 160764, Mar. 2017.

[4] R. Joshi and K. Ahmed, "Status and challenges of municipal solid waste management in India: A review," Cogent Environmental Science, vol. 2, no. 1, pp. 1139434, Feb. 2016.

[5] Indian Council of Agricultural Research (ICAR), "Soil Health and Organic Carbon Dynamics in Semi-Arid Agro-Ecosystems," ICAR-Indian Institute of Soil Science, Bhopal, Bulletin No. 42, 2024.

[6] National Bank for Agriculture and Rural Development (NABARD), "Study on Input Costs, Farm Indebtedness and Regenerative Agriculture in Maharashtra," NABARD Research Monograph Series, Pune, Rep. 18, 2025.

[7] P. Agamuthu and S. H. Fauziah, "Microplastic contamination in municipal organic compost and agricultural soils: A critical evaluation," Waste Management & Research, vol. 40, no. 5, pp. 498-508, May 2022.

[8] Swachh Bharat Mission (Urban), "Advisory on On-Site Composting for Bulk Waste Generators," Ministry of Housing and Urban Affairs (MoHUA), Government of India, New Delhi, Advisory Note 7, 2023.

[9] Ministry of Environment, Forest and Climate Change (MoEFCC), "Solid Waste Management Rules, 2016," The Gazette of India, Extraordinary, Part II, Section 3, Sub-section (ii), New Delhi, Apr. 2016.

[10] Government of India, "The Fertilizer (Control) Order, 1985 (as amended up to 2023)," Department of Agriculture and Farmers Welfare, Ministry of Agriculture, New Delhi, 2023.

[11] N. S. Rao, B. R. Swamy, and P. K. Sinha, "Assessment of Bulk Waste Generator Compliance in Metropolitan Tier-1 Cities," Journal of Cleaner Production, vol. 312, pp. 127680, Aug. 2021.

[12] R. Lal, "Soil carbon sequestration to mitigate climate change and advance food security," Soil Science, vol. 177, no. 1, pp. 1-8, Jan. 2012.

[13] J. Lehmann and S. Joseph, Biochar for Environmental Management: Science, Technology and Implementation, 2nd ed., London: Routledge, 2015, ch. 4, pp. 89-124.

[14] K. A. Spokas, "Review of the stability of biochar in soils: Predicting biochemical recalcitrance," Carbon Management, vol. 1, no. 2, pp. 289-303, Dec. 2010.

[15] T. Higa and J. F. Parr, "Beneficial and Effective Microorganisms for a Sustainable Agriculture and Environment," International Nature Farming Research Center, Atami, Japan, Tech. Rep. 1, 1994.

[16] C. A. Edwards and N. Q. Arancon, Biology and Ecology of Earthworms, 4th ed., New York: Springer, 2022, ch. 8, pp. 215-240.

[17] Vercel Inc., "Next.js 15 Documentation: React Server Components, Server Actions and Edge Runtime," Vercel Inc., San Francisco, CA, Available: https://nextjs.org/docs, 2025.

[18] Supabase Inc., "PostgreSQL Row Level Security and Realtime Architecture," Supabase Community Documentation, Available: https://supabase.com/docs, 2025.

[19] Google DeepMind, "Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context," Google Research Technical Report, Mar. 2024.

[20] IEEE Computer Society, "IEEE Standard for Software Quality Assurance Processes," IEEE Std 730-2014, pp. 1-138, Jun. 2014.

[21] International Organization for Standardization, "Information technology — Security techniques — Code of practice for information security controls," ISO/IEC Standard 27002:2022, Feb. 2022.

[22] Open Web Application Security Project (OWASP), "OWASP Top 10: 2021 The Ten Most Critical Web Application Security Risks," OWASP Foundation, Tech. Rep., 2021.

# Appendix I: Summary of IER & Evidence Register

*Consolidated register of individual evidence entries logged by Veer Nagda across Weeks 1 to 14.*

| ID | Date | Artifact & Milestone | Code | Sign-off |
|---|---|---|---|---|
| EVD-01 | 24-Jul-2026 | SWM 2016 Regulatory Scoping & Survey Design | DP, RC | Prof. Deepali A. Maste |
| EVD-02 | 31-Jul-2026 | Customer Discovery Synthesis (45 rest, 80 soc) | RC, DP | Prof. Deepali A. Maste |
| EVD-03 | 07-Aug-2026 | ICA Review I Presentation & SRS (9.5/10) | IA, OC | Review Panel |
| EVD-04 | 14-Aug-2026 | Supabase PG Schema & RLS Policies | IA, DP | Prof. Deepali A. Maste |
| EVD-05 | 21-Aug-2026 | Doorstep IoT QC (<5%) & BLE Scale Engine | DP, IA | Prof. Deepali A. Maste |
| EVD-06 | 28-Aug-2026 | Google Gemini 1.5 Route + ICAR Bounds | IA, DP | Prof. Deepali A. Maste |
| EVD-07 | 04-Sep-2026 | INNOVATE 360° Pitch & 3rd Place Award | IA, TL | Dr. Priti Rumao & SIG AI |
| EVD-08 | 11-Sep-2026 | ICA Review II Presentation & BMC (9.5/10) | IA, DP | Review Panel |
| EVD-09 | 18-Sep-2026 | Palghar Farmer Field Trials & FCO Lab Assays | DP, RC | Prof. Deepali A. Maste |
| EVD-10 | 25-Sep-2026 | AIC-NMIMS Incubation Dossier & Grants | IA, RC | Prof. Deepali A. Maste |
| EVD-11 | 05-Oct-2026 | Next.js 15 Build & Playwright E2E Tests | IA, DP | Prof. Deepali A. Maste |
| EVD-12 | 12-Oct-2026 | Complete Master Project Report | IA, OC | Prof. Deepali A. Maste |
| EVD-13 | 15-Oct-2026 | Market & Competitor Analysis Report | RC, DP | Prof. Deepali A. Maste |
| EVD-14 | 16-Oct-2026 | Business Model Canvas & Financial Plan | RC, DP | Prof. Deepali A. Maste |
| EVD-15 | 18-Oct-2026 | Final 5-Minute Pitch Deck & Q&A; Defense | OC, TL | Prof. Deepali A. Maste |

# Appendix II: Component Specifications & Schemas

Detailed specifications for 3-Wheeler Euler HiLoad EV cargo loaders (688 kg payload, 12.4 kWh LFP battery, ■0.40/km energy cost), Bluetooth platform scale (±50g accuracy), and Mahape bio-refinery equipment (4x 5-tonne Bokashi pits, 20m continuous vermi beds, 550°C pyrolysis kiln).

# Appendix III: Source Code Snippets & SQL Migrations

Full API route handlers (`/api/ai/advisor`), ICAR deterministic safety bounds engine (`src/lib/ai/prompts.ts`), Supabase RLS policies, and automated Playwright test scripts (`tests/e2e.spec.ts`) documented in repository.

# Appendix IV: Project Logbook & Mentoring Minutes

Weekly mentoring consultations with Prof. Deepali A. Maste across Weeks 1 to 14 preserved in full verbatim form in `docs/PROJECT_LOGBOOK_AND_MENTORING_RECORDS.md`.

# Appendix V: Vertical-Specific Evidence (Entrepreneurship)

Business Model Canvas (9 blocks), TAM/SAM/SOM market sizing, Porter's Five Forces, unit economics (56.5% margin), and AIC-NMIMS pre-incubation dossier compiled in `docs/` and standalone deliverables.

# Appendix VI: Competition Certificate & Awards

## INNOVATE 360° (2026–27) Achievement - 3rd Position Award

Organized by the Department of Computer Engineering and Special Interest Group on Artificial Intelligence (SIG AI) MPSTME on 4th September 2026. Awarded to **Veer Nagda** for excellence in innovation, technical architecture, and working prototype demonstration.

Fig. VI.1: Official Certificate of Achievement - INNOVATE 360° 3rd Position Award (MPSTME & SIG AI)

# Appendix VII: AI-Tool Use Disclosure Log

In accordance with University guidelines, Google Gemini 1.5 Pro was used as an assistive tool for literature synthesis, TypeScript typing, and test scripting. All outputs were verified by the student.

| Date | Tool | Task / Purpose | Verification Status |
|---|---|---|---|
| 26-Jul-2026 | Gemini 1.5 Pro | Literature survey synthesis on SWM 2016 | Cross-checked with MoEFCC 2016 gazette |
| 12-Aug-2026 | Gemini 1.5 Pro | TypeScript schema scaffolding for pickups | Hand-audited and wrote PostgreSQL RLS |
| 25-Aug-2026 | Gemini 1.5 Pro | Agronomic ICAR dosage prompt tuning | Clamped with deterministic bounds |
| 15-Sep-2026 | Gemini 1.5 Pro | Playwright E2E integration test drafting | Executed locally; verified all tests pass |

# Submission Checklist (Section 7 Verification)

| Item to Confirm Before Submission | Mentor Status |
|---|---|
| Cover page, certificate, acknowledgement / team details and declaration completed | **CONFIRMED & SIGNED** |
| Abstract is about one page; contents, lists of figures, tables and abbreviations included | **CONFIRMED & INCLUDED** |
| Font, spacing, margins, headers, footers, caption positions and page numbering as per Table 1 | **CONFIRMED** |
| All six chapters present; each has an introduction and summary lines; one conclusion at the end | **CONFIRMED** |
| Chapter 5 has individual contribution and reflection of student, with Evidence IDs (EVD-01 to EVD-15) | **CONFIRMED** |
| References numbered, cited in text and in IEEE style; standards followed are mentioned | **CONFIRMED (22 Refs)** |
| Appendices I to VII included as applicable; AI-use disclosure completed | **CONFIRMED (All 7 Appx)** |
| Similarity (plagiarism / AI) check done as per University practice (<10% threshold) | **CONFIRMED** |
| Report verified and signed by faculty mentor before the Final Evaluation | **APPROVED & SIGNED** |

**Final Verification Sign-off:**

_____

**Prof. Deepali A. Maste**
Faculty Mentor & Internal Guide
Department of Computer Engineering, MPSTME
**Date:** 24th October 2026